

Meander Line Antenna Simulation

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Calculations based on paper

$$d = 0.16\lambda_g$$

$$s = 0.41\lambda_g$$

$$L = 0.70\lambda_g$$

$$w = 0.05\lambda_g$$

$$\lambda_g = \frac{\lambda}{\sqrt{\epsilon_{ff}}}$$

$$\epsilon_{ff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 10\frac{h}{w}\right)^{-1/2}$$

Assuming a standard 4-layer PCB:

$$h = 1.6 \text{ mm}, \quad \epsilon_r = 4.5$$

First we calculate ϵ_{ff} to get λ_g :

$$\lambda_g = \frac{\lambda}{\sqrt{\epsilon_{ff}}},$$

$$\lambda = \frac{c}{f} = \frac{3 \cdot 10^8}{1060 \text{ MHz}} = 28.3 \text{ cm},$$

$$\epsilon_r = 4.5, \quad h = 1.6 \text{ mm},$$

$$\therefore \epsilon_{ff} = 3.73$$

Therefore:

$$\lambda_g = 0.1465 \text{ m} = 14.65 \text{ cm}$$

The paper mentions a necessary dimension correction ΔL :

$$\Delta L = 0.5w, \quad w = \frac{c}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}}$$

The meander line dimensions before corrections are:

$$\begin{aligned}d &= 0.16\lambda_g = 0.02344 \text{ m} \\s &= 0.42\lambda_g = 0.06153 \text{ m} \\L &= 0.70\lambda_g = 0.10255 \text{ m} \\w &= 0.05\lambda_g = 0.007325 \text{ m}\end{aligned}$$

Now we calculate ΔL :

$$\Delta L = 0.5 \left(\frac{3 \cdot 10^8 \text{ m/s}}{2 \cdot 1.06 \text{ GHz}} \right) \sqrt{\frac{2}{4.5 + 1}} = 0.04266 \text{ m}$$

The corrected values are:

$$\begin{aligned}S_p &= s + \Delta L = 0.10419 \text{ m} \\d_p &= d + \Delta L = 0.0661 \text{ m} \\L_p &= L + \Delta L = 0.14521 \text{ m}\end{aligned}$$

Now we calculate the HPBW and the directivity:

$$k_0 = \frac{\pi}{\sqrt{\epsilon_{ff}} L_p} = 11.20208$$

$$\begin{aligned}\theta_E &= 2.45 \text{ rad} = 140^\circ \\ \theta_H &= 1.867 \text{ rad} = 107^\circ\end{aligned}$$

Therefore:

$$D = \frac{41253}{\theta_E \theta_H} = 2.754 \approx 4.4 \text{ dBi}$$

Simulation Results

The antenna was simulated using the following functions:

1. singleDipole() → This function calculates the E- and H-field contributions from a single Hertzian dipole at a single point in space
2. oneHertzianAllPoints() → This function calculates the E- and H-field contributions from a single Hertzian dipole to all points in space
3. fullTraceAllPoints() → This function calculates the E- and H-field contributions from a whole meander line trace to all points in space
4. MeanderTotalRadiation() → This function calculates the E- and H-field contributions of all Meander traces for all points in space
5. For the field contributions of each dipole the current distribution is assumed sinusoidal with variations based on $I_0 e^{-j\beta l}$

The results were:

$$E_{HPBW} = 117.30^\circ, \quad H_{HPBW} = 151.68^\circ$$

Which gives a directivity of:

$$D = \frac{41253}{117.3^\circ \times 151.68^\circ} = 2.3184 = 3.65 \text{ dBi}$$

Conclusion

There was a discrepancy between calculations and simulations the directivity of:

$$4.4 \text{ dBi} - 3.65 \text{ dBi} = 0.75 \text{ dBi}$$

The radiation pattern can be seen by opening `Meander_Line_Radiation_Pattern_3D.html` or by running the `Antenna_sim.ipynb`